ii) Measurement of large distances

For measuring larger distances such as the height of a tree, distance of the Moon or a planet from the Earth, some special methods are adopted. Triangulation method, parallax method and radar method are used to determine very large distances.

Triangulation method for the height of an accessible object

Let AB = h be the height of the tree or tower to be measured. Let C be the point of observation at distance x from B. Place a range finder at C and measure the angle of elevation, $\angle ACB = \theta$ as shown in Figure 1.3.

From right angled triangle ABC,

$$\tan\theta = \frac{AB}{BC} = \frac{h}{x}$$

(or)

$$\text{height } h = x \tan\theta$$

Knowing the distance x, the height h can be determined.

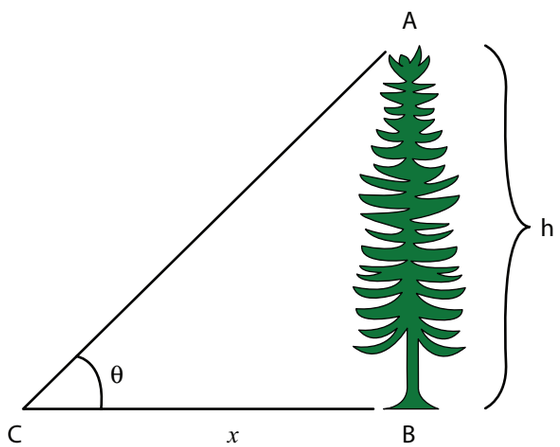


Figure 1.3 Triangulation method

Range and order of lengths of various objects are listed in Table 1.5

EXAMPLE 1.1

From a point on the ground, the top of a tree is seen to have an angle of elevation 60°. The distance between the tree and a point is 50 m. Calculate the height of the tree?

Solution

$$\text{Angle } \theta = 60°$$

The distance between the tree and a point $x = 50$ m

$$\text{Height of the tree (h)} = ?$$

$$\text{For triangulation method } \tan\theta = \frac{h}{x}$$

$$h = x \tan\theta$$
$$= 50 \times \tan 60°$$
$$= 50 \times 1.732$$
$$h = 86.6 \text{ m}$$

The height of the tree is 86.6 m.

Parallax method

Very large distances, such as the distance of a planet or a star from the Earth can be measured by the parallax method. *Parallax is the name given to the apparent change in the position of an object with respect to the background, when the object is seen from two different positions*. The distance between the two positions (i.e., points of observation) is called the basis (b). Consider any object at the location O (Figure 1.4)

Let L and R represent the positions of the left and right eyes of the observer respectively.

The object (O) is viewed with the left eye (L) keeping the right eye closed and the same object (O) is viewed with the right eye (R) keeping the left eye closed.

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